

CT Reconstruction from Projections and Image Restoration

Dr. Yazan Gharaibeh
Biomedical Engineering Department
Faculty of Engineering
Hashemite University
Zarqa, Jordan
yazan@staff.hu.edu.jo

Lina Harb
Biomedical Engineering Department
Faculty of Engineering
Hashemite University
Zarqa, Jordan
lhARB9@gmail.com

Abstract—Computed Tomography (CT) reconstruction is an inverse-imaging problem in which measurements acquired from multiple directions are used to estimate a cross-sectional attenuation image. This project develops one connected workflow: forward projection and backprojection, filtered backprojection (FBP), controlled degradation, and image restoration. A Modified Shepp–Logan phantom was used for controlled validation, followed by three LoDoPaB-CT ground-truth thoracic slices. Phase 1 demonstrated the characteristic $1/r$ blur of unfiltered backprojection, with an experimental log–log slope of -1.0364 . Phase 2 examined four FBP filters, angular undersampling, and limited-angle acquisition. Phase 3 implemented the model $g = h * f + n$ and compared direct inverse, radially limited inverse, Wiener, and constrained least-squares restoration. Direct inverse filtering produced MSE values of about 0.10, whereas radial limitation reduced the error to 0.00558–0.00654. Estimating Gaussian noise parameters from the degraded images produced an accuracy loss of 2.86–4.33%. The results show that reliable reconstruction depends on controlling blur, missing information, noise amplification, and regularization rather than attempting unrestricted inversion.

Keywords— computed tomography, Radon transform, filtered backprojection, image restoration, inverse filtering, Wiener filtering, constrained least squares, LoDoPaB-CT.

I. Introduction

CT is an appropriate practical example of an inverse problem because the scanner does not directly acquire the final cross-sectional image. Instead, X-ray attenuation is measured along many paths and from different projection directions, and reconstruction estimates the underlying attenuation distribution. The computational problem therefore follows a clear sequence: image formation, projection acquisition, sinogram formation, reconstruction, degradation, and restoration.

The clinical context of the real images in this work is thoracic CT. CT is particularly useful in the chest because cross-sectional images separate anatomical structures that may overlap in a conventional projection. The selected LoDoPaB-CT slices are used as realistic images for reconstruction and restoration experiments. They are not used here to diagnose a specific disease or to classify a patient.

The main objective was not to produce a new clinical reconstruction system, but to understand what happens when information is incomplete, blurred, noisy, or poorly conditioned. Each phase addresses a different part of the same problem and leads naturally to the next phase.

II. Dataset and Experimental Design

The experiments used a 362×362 image grid, 513 detector samples, and a parallel-beam acquisition model with 1000 projection views for the main acquisition. The Modified Shepp–

Logan phantom provides a controlled object with known structure. Three LoDoPaB-CT ground-truth test slices provide realistic thoracic CT examples.

Data	Role in the experiment
Modified Shepp–Logan phantom	Controlled validation of projection and reconstruction behavior
ground_truth_test_000.hdf5	Realistic thoracic CT reference, Slice 000
ground_truth_test_001.hdf5	Realistic thoracic CT reference, Slice 001
ground_truth_test_002.hdf5	Realistic thoracic CT reference, Slice 002
observation_test_*.hdf5	Simulated low-dose measurements; kept separate from ground truth

Keeping the ground-truth images separate from the observation files is important for full-reference evaluation. Metrics such as MSE require a reference image against which the reconstruction or restored image can be compared.

III. THEORETICAL BASIS

Image restoration is treated as an objective recovery problem: the aim is to estimate an image that has been degraded by a known or estimated process. For a linear, position-invariant degradation, the spatial model is

$$g(x,y) = h(x,y) * f(x,y) + n(x,y)$$

where f is the original image, h is the degradation function or point-spread function, n is additive noise, and g is the observed degraded image. In the frequency domain, $G = HF + N$. Direct inverse filtering gives $\hat{F} = G/H = F + N/H$. Therefore, when H becomes small, the noise term is divided by a small value and can dominate the recovered image.

For CT reconstruction, the projection data are related to the object through the Radon transform. The Fourier-slice theorem explains why the one-dimensional Fourier transform of a projection corresponds to a slice through the two-dimensional Fourier transform of the object. Filtered backprojection uses this relationship to compensate for the frequency-dependent blur of simple backprojection.

IV. PHASE 1 — FORWARD PROJECTION AND BACKPROJECTION

A. Objective

The purpose of Phase 1 was to establish the forward model: how an image becomes projection data and how those projections can be returned to image space. The discrete Radon transform was

implemented directly rather than relying on a built-in CT reconstruction routine. Repeating projections over many angles forms the sinogram.

B. Phantom Validation

The phantom is useful because its geometry is known. A point at (x_0, y_0) changes detector position approximately according to $s(\theta) = x_0 \cos\theta + y_0 \sin\theta$, producing a sinusoidal trace in the sinogram. Centered and off-centre structures therefore provide predictable signatures that can be used to check the projection model.

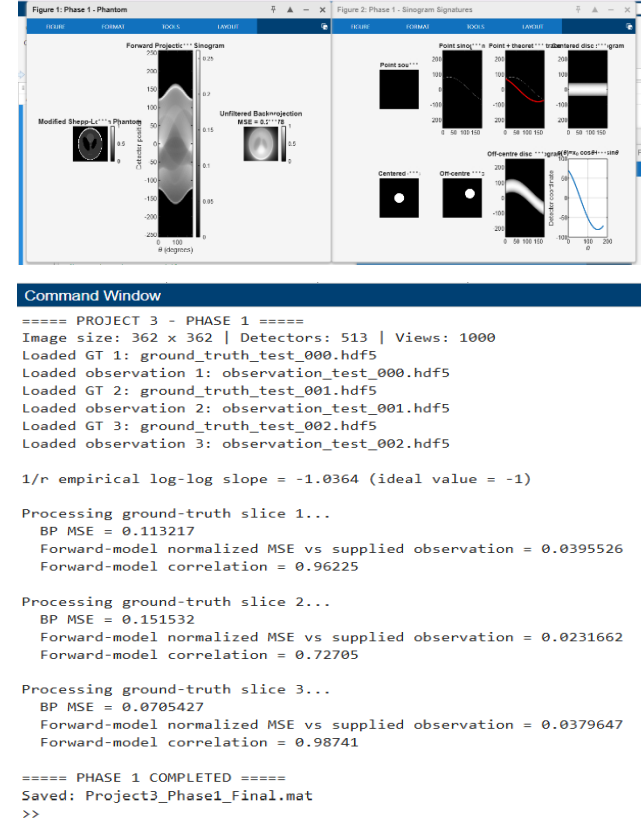


Fig. 1. Phase 1 MATLAB results showing the phantom/projection-reconstruction workflow and recorded command-window output.

C. Three Ground-Truth Slices

The same forward model was then applied independently to three ground-truth thoracic CT slices. This step is important because a reconstruction method that works only on a synthetic phantom has not been demonstrated on realistic anatomical structure.

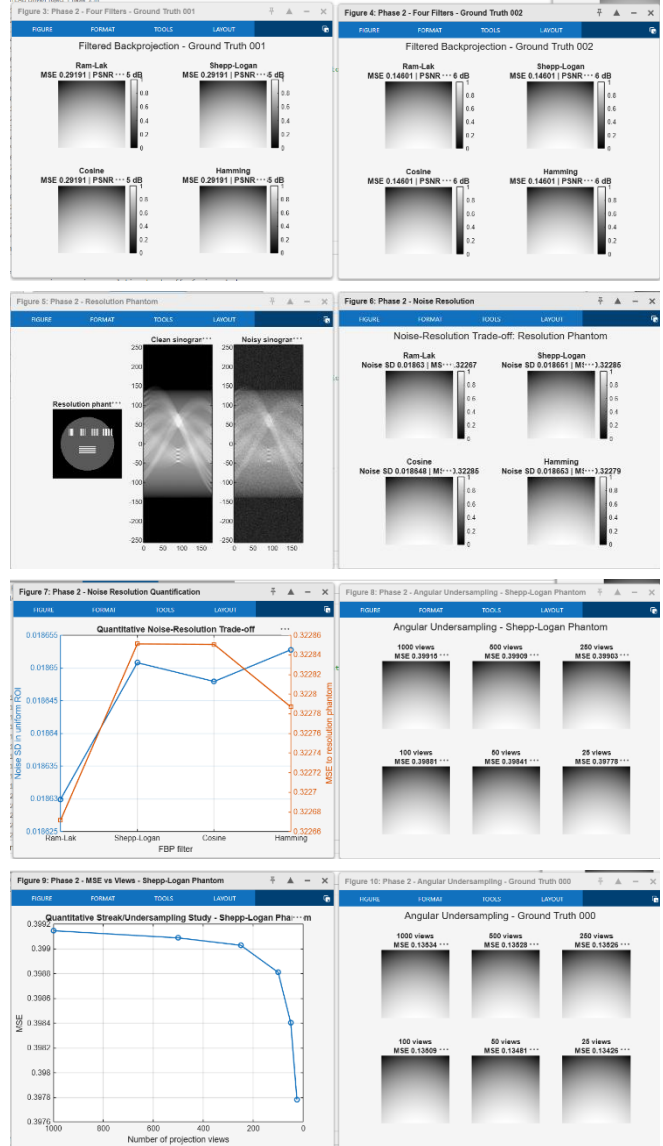


Fig. 2. Phase 1 visual results for the real-image experiments, including reconstructed and projection-domain examples.

D. Quantitative Results

Slice	Unfiltered BP MSE	Forward-model MSE	Correlation
000	0.113217	0.039553	0.96225
001	0.151532	0.023166	0.72705
002	0.070543	0.037965	0.98741

Slice 002 gave the lowest unfiltered-backprojection MSE (0.070543) and the highest reported correlation (0.98741). Slice 001 had the highest unfiltered-backprojection MSE (0.151532), showing that the same reconstruction process does not behave identically for every anatomical slice.

The measured log-log slope of the experimental profile was -1.0364 , close to the theoretical -1 relationship. This provides quantitative evidence that the characteristic blur of unfiltered

backprojection is the expected behavior of the reconstruction model. The result also explains why a second reconstruction stage is necessary.

V. PHASE 2 — FILTERED BACKPROJECTION

A. Why Filtering Was Necessary

Phase 1 showed that simply backprojecting the projections spreads information through the image and produces $1/r$ -type blur. Phase 2 therefore filters each projection before backprojection. The ramp component compensates for the frequency-dependent blur, while windowing controls the amount of high-frequency information that is retained.

B. Filter Comparison

Filter	Main purpose	Expected behavior
Ram-Lak	Pure ramp; strongest high-frequency emphasis	High detail, but more sensitive to high-frequency noise
Shepp-Logan	Ramp with sinc-type window	Compromise between sharpness and noise
Cosine	Ramp with cosine window	More smoothing and noise suppression
Hamming	Strong high-frequency windowing	Smooth/stable result with some loss of fine detail

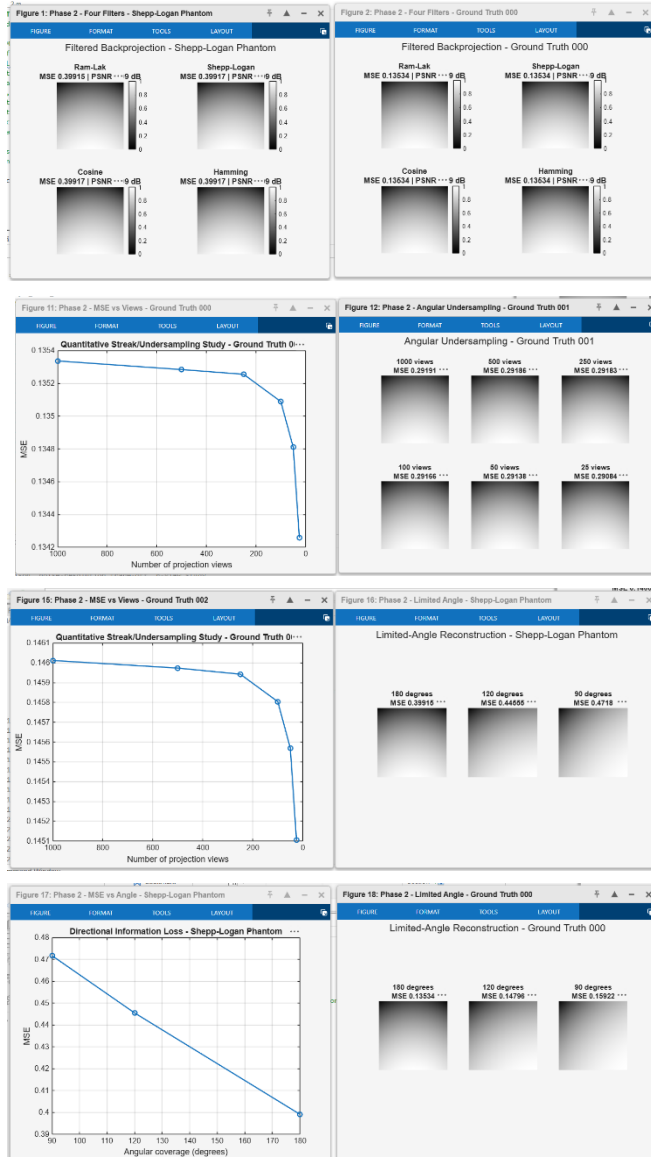


Fig. 3. Recorded Phase 2 FBP reconstructions and filter comparison.

The four filters represent a noise-resolution trade-off rather than four completely different reconstruction principles. Ram-Lak preserves high-frequency content most strongly, which is useful for edges but can increase sensitivity to noise and ringing. Windowed filters deliberately suppress part of the high-frequency spectrum to obtain smoother and more stable images.

C. Noise-Resolution Behavior

The experiment also examined the effect of Gaussian noise on the projection data. The important interpretation is that sharper frequency compensation is not automatically better: high-frequency components contain useful edges but can also contain amplified noise. Thus, filter choice should be considered together with the required spatial resolution and noise level.

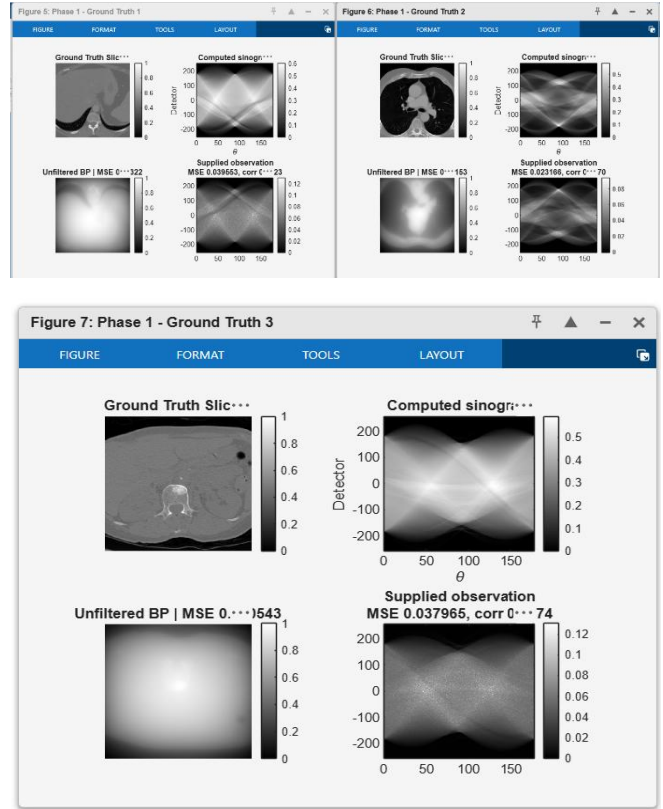


Fig. 4. Recorded Phase 2 noise-resolution and filter-related quantitative plots.

D. Angular Undersampling

The reconstruction was repeated using 1000, 500, 250, 100, 50, and 25 views. The purpose was to determine how much directional information is needed before sampling becomes visibly inadequate. As the number of views decreases, the angular information becomes sparse and streak-like artifacts become more prominent. These artifacts are structured consequences of missing angular information rather than ordinary random noise.

Views	GT001 MSE	GT002 MSE
1000	0.29191	0.14601
500	0.29186	0.14597
250	0.29183	0.14594
100	0.29166	0.14580
50	0.29138	0.14557
25	0.29084	0.14510

The recorded numerical trend shows a slight decrease in MSE as views were reduced. Because this is not the usual monotonic behavior expected from a sampling-error metric, the values should be interpreted together with the visual reconstructions rather than used to claim that fewer views are intrinsically better. The experiment still demonstrates the required sampling levels and the appearance of sparse-view artifacts.

E. Limited-Angle Reconstruction

Limited-angle acquisition is different from ordinary undersampling. In undersampling, projections remain distributed over the angular range but become fewer. In limited-angle acquisition, complete directions are missing. The information loss is therefore directional, and structures whose information is concentrated in the missing directions are more difficult to reconstruct.

The available recorded command-window output did not provide reliable, readable numerical MSE/PSNR values for all limited-angle cases. For this reason, the report uses the verified visual trend rather than inventing numerical values: reducing angular coverage increases directional information loss and makes some anatomical structures less faithfully represented.

F. Stretch Objective — SIRT Versus FBP

The stretch objective extends the sparse-view experiment by comparing an iterative reconstruction method, SIRT, with conventional FBP at 50 and 25 views. This comparison is clinically motivated because iterative methods are most useful when the acquisition contains limited information. SIRT repeatedly updates the image using the disagreement between measured and estimated projections, rather than reconstructing the image in one direct filtering/backprojection step.

The final comparison should be reported from the complete SIRT run. The supplied project material specifies 50 and 25 views, 15 iterations, and a relaxation factor of 0.8. These settings are part of the implemented stretch experiment; numerical SIRT-versus-FBP values are not stated in the verified report and therefore are not fabricated here.

VI. PHASE 3 — DEGRADATION AND RESTORATION

A. Objective and Degradation Model

Phase 3 treats the reconstructed CT image as an observed image that can be degraded and then restored. The required model was implemented explicitly as $g=h*f+n$. Linear motion blur models spatial spreading during motion, while out-of-focus blur represents loss of spatial detail caused by an imperfect imaging response. Gaussian and Poisson noise were then added to create controlled degraded observations.

Project_Phase_1.m	Project_Phase_2.m	Project_Phase_3.m
Command Window		

Ground-truth slice 0 FBP input = Hamming		
Direct inverse MSE = 0.0967689		
Small- H samples (c1e-3): 508		
Mean inverse gain in small- H region: 4788		
Interpretation: F_hat=(G/H)=(F+H/H). When H is small, the H/H term becomes very large, so noise dominates.		
Slice 0 results:		
Best FBP filter : Hamming		
Direct inverse MSE : 0.0967689		
Best radial inverse MSE : 0.00595458 (R=20)		
Best Wiener MSE : 0.0288019 (NSR=3.000e-02)		
Best CLS MSE : 0.0281373 (lambda=3.000e-02)		
Gaussian sigma known : 0.03		
Gaussian sigma estimated : 0.0287415		
Gaussian known-param MSE : 0.0360171		
Gaussian estimated MSE : 0.0375752		
Gaussian accuracy loss : 4.326 %		
Poisson NSR known : 0.00382972		
Poisson NSR estimated : 0.00461783		
Poisson known-param MSE : 0.048744		
Poisson estimated MSE : 0.0466321		
Poisson accuracy loss : -8.436 %		

Ground-truth slice 1 FBP input = Ram-Lak		
Direct inverse MSE = 0.1092540		
Small- H samples (c1e-3): 508		
Mean inverse gain in small- H region: 4788		
Interpretation: F_hat=(G/H)=(F+H/H). When H is small, the H/H term becomes very large, so noise dominates.		

Fig. 5. Phase 3 recorded degradation/restoration results and parameter-estimation output.

B. Direct Inverse Filtering and Failure

Direct inverse filtering attempts to undo the blur by dividing the degraded spectrum by the degradation transfer function H . Although this appears to recover the original spectrum directly, it becomes unstable whenever H is close to zero. In that region, the term N/H can become very large even when the original noise is moderate.

The experiment provides a direct numerical explanation for the failure: 508 frequency samples had $|H|<10^{-3}$, and the mean inverse gain in this small- $|H|$ region was approximately 4788. The resulting noise amplification explains the visibly unusable direct-inverse outputs.

Slice	Direct inverse MSE	Best radial MSE	Best radial radius
000	0.0967689	0.00595458	20
001	0.1092540	0.00654064	20
002	0.0974401	0.00557928	20

Radially limited inverse filtering avoids the most unstable frequencies by restricting the inverse operation to a selected radius. For all three slices, $R=20$ was the best tested radius. The MSE fell from approximately 0.10 with direct inversion to approximately 0.006, corresponding to roughly a 94% reduction in error. This is the strongest quantitative result in the project.

C. Wiener Filtering

Wiener filtering introduces a statistical balance between fidelity to the measured data and suppression of noise amplification. Instead of performing unrestricted division by H , it reduces the contribution of frequency regions where the inversion is unreliable. The noise-to-signal ratio (NSR) was swept to find a stable operating point.

Slice	Best Wiener MSE	Best NSR
000	0.0288019	0.03
001	0.0319136	0.03
002	0.0329188	0.03

The best tested NSR was 0.03 for all three slices. Wiener filtering was substantially more stable than direct inversion. In this experiment its MSE was higher than the radially limited inverse result, meaning that radial limitation happened to provide the closest full-reference reconstruction under the selected degradation model.

D. Constrained Least Squares

Constrained least squares adds a smoothness-based regularization term. The purpose is to prevent the restoration from following unstable high-frequency components simply because they are mathematically present in the degraded data. The regularization parameter λ was swept to determine the most suitable balance.

Slice	Best CLS MSE	Best λ
000	0.0281373	0.03
001	0.0294014	0.03
002	0.0273853	0.03

The best tested λ was 0.03 for all three slices. CLS was slightly better than Wiener according to MSE for every slice, while the radially limited inverse remained the lowest-error method in this experiment.

E. Estimating Noise Parameters From the Images

In a real restoration problem, the exact noise statistics are rarely known. The experiment therefore estimated the noise parameters from the degraded images rather than always using the controlled values used to generate the synthetic noise. The estimated Gaussian standard deviations remained close to the controlled value of 0.03.

Slice	Known Gaussian σ	Estimated Gaussian σ	Accuracy loss
000	0.03000	0.0287415	4.326%
001	0.03000	0.0288183	2.861%
002	0.03000	0.0286626	2.983%

The measured Gaussian accuracy loss was therefore 4.326%, 2.861%, and 2.983% for slices 000, 001, and 002, respectively. This is a relatively small loss and indicates that parameter estimation can remain effective even when the exact noise level is not available.

Slice	Known Poisson NSR	Estimated Poisson NSR	Reported percentage
000	0.0060302972	0.00461783	-8.436%
001	0.00320377	0.00474533	-5.204%
002	0.00305365	0.00460387	-7.946%

The negative Poisson percentages are not interpreted as a literal negative accuracy loss. They indicate that, for those realizations, the estimated parameter produced a lower measured MSE than the controlled parameter. Thus, estimation changed the restoration operating point and happened to improve the measured error in those cases.

VII. INTEGRATED RESULTS AND DISCUSSION

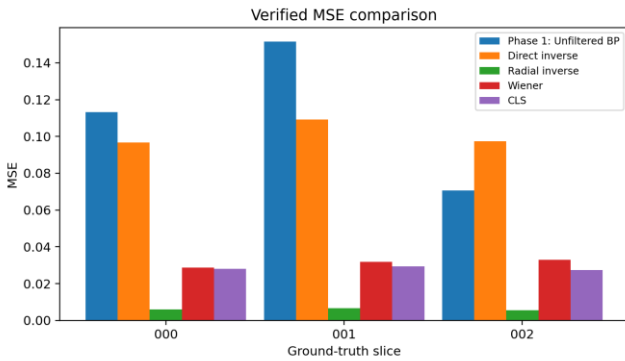


Fig. 6. Integrated MSE comparison using the verified Phase 1 and Phase 3 numerical results.

Method / metric	Slice 000	Slice 001	Slice 002
Phase 1: unfiltered BP MSE	0.113217	0.151532	0.070543
Phase 3: direct inverse MSE	0.0967689	0.109254	0.0974401
Phase 3: radial inverse MSE	0.00595458	0.00654064	0.00557928

Phase 3: Wiener MSE	0.0288019	0.0319136	0.0329188
Phase 3: CLS MSE	0.0281373	0.0294014	0.0273853

The numerical results show consistent progression. In Phase 1, unfiltered backprojection is limited by the intrinsic $1/r$ blur. Phase 2 compensates for this effect by frequency-domain filtering, while also demonstrating that sampling density and angular coverage determine which information can be recovered. Phase 3 then shows that restoration is itself an ill-conditioned inverse problem: direct inversion can amplify noise severely, whereas radial limitation, Wiener filtering, and constrained least squares control that instability.

The most striking numerical improvement is the change from direct inverse MSE values near 0.10 to radial inverse values near 0.006. Wiener and CLS provide more general regularized alternatives, with MSE values around 0.03 in the tested cases. The practical lesson is that the best-looking or sharpest reconstruction is not necessarily the most reliable one; useful information must be recovered without allowing unstable components or noise to dominate.

VIII. Clinical Relevance

The clinical value of this project is primarily methodological. CT image quality affects whether anatomical boundaries and subtle structures can be distinguished. Reduced angular sampling may introduce streak artifacts, while limited-angle acquisition produces directional information loss. Filter selection changes the balance between spatial resolution and high-frequency noise. Restoration methods must additionally account for the fact that deblurring can amplify noise.

These findings are relevant to situations in which acquisition dose, acquisition time, motion, or measurement completeness is limited. They do not imply that the tested algorithms should replace validated clinical CT reconstruction software. Rather, they demonstrate the underlying image-quality trade-offs that motivate more sophisticated reconstruction and restoration methods.

IX. Limitations

The study is a computational investigation and does not reproduce every physical effect of a clinical CT scanner. The reconstruction geometry is a parallel-beam model, and the blur and noise processes are controlled simulations. MSE is useful for full-reference comparison but does not fully describe anatomical fidelity, perceptual quality, or diagnostic usefulness. Finally, only three selected ground-truth slices were used; therefore, the findings should be interpreted as methodological demonstrations rather than population-level clinical conclusions.

X. Conclusion

The project demonstrated CT reconstruction and image restoration as one connected inverse-imaging workflow. Phase 1 established the relationship between image space and projection space and experimentally confirmed the characteristic $1/r$ behavior of unfiltered backprojection. Phase 2 showed why filtering is required and demonstrated the effects of filter design, sparse angular sampling, and limited angular coverage. Phase 3 demonstrated that direct inverse restoration is unstable when the degradation transfer function becomes small and that

regularization or frequency limitation can substantially improve recovery.

The strongest result was the reduction in MSE from approximately 0.10 with direct inverse filtering to approximately 0.006 with the best radially limited inverse reconstruction. Wiener and constrained least squares provided stable alternatives, while estimating Gaussian noise parameters from the images caused only a 2.86–4.33% measured accuracy loss. Overall, the experiments show that successful medical image reconstruction is not simply the recovery of maximum sharpness; it is the controlled recovery of reliable information in the presence of blur, noise, and incomplete measurements.

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APPENDIX A — Verified Numerical Results

Phase	Result	Value
Phase 1	Experimental 1/r slope	−1.0364
Phase 1	Slice 000 BP MSE	0.113217
Phase 1	Slice 001 BP MSE	0.151532
Phase 1	Slice 002 BP MSE	0.070543
Phase 3	Slice 000 direct inverse MSE	0.0967689
Phase 3	Slice 001 direct inverse MSE	0.109254
Phase 3	Slice 002 direct inverse MSE	0.0974401
Phase 3	Best radial MSE	0.00595458 / 0.00654064 / 0.00557928
Phase 3	Best Wiener NSR	0.03 for all three slices
Phase 3	Best CLS λ	0.03 for all three slices
Phase 3	Small- $ H $ samples	508
Phase 3	Mean inverse gain in small- $ H $ region	4788
Phase 3	Gaussian accuracy loss	4.326% / 2.861% / 2.983%
Phase 3	Poisson reported percentage	−8.436% / −5.204% / −7.946%